

RAHEL RICKENBACH

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*Final-year PhD candidate working on **inverse reinforcement learning** and **learning from demonstrations** under noisy, suboptimal, and safety-critical data, and on integrating these with **foundation models** for language-based robot control and **LLM** fine-tuning. I further use **control theory** to analyze and improve **learning-based methods**, with an emphasis on interpretability. I am seeking a research internship applying **machine learning, reinforcement learning, and foundation models** to challenging real-world problems.*

EXPERTISE

- **Machine learning** (supervised learning, reinforcement learning, structured state-space models)
- **Learning from demonstrations** (inverse reinforcement learning, imitation learning)
- **Robotics and control** (safe optimization-based control, state estimation, motion planning)

EXPERIENCE

PhD Candidate
ETH, advised by Prof. M. Zeilinger

Zurich, Switzerland
August 2022 - today

- Analyzed the memory behaviour of deep sequence models (state-space and attention-based) using control-theoretic tools. **Tech:** PyTorch, JAX
- Designed and experimentally validated an imitation learning pipeline compressing perceived context into a constraint-aware controller with generalization capabilities. **Tech:** Python, C++, ROS, CasADi
- Developed and experimentally validated a language-to-control method with inference-time hallucination detection and capability extension via learning from demonstrations. **Tech:** Python, JAX, ROS, PandaGym
- Established theoretical foundations for inverse reinforcement learning from noisy, suboptimal, and safety-critical demonstrations. **Tech:** Python

Research Assistant
ETH

Zurich, Switzerland
November 2021 - April 2022

- Developed and experimentally validated a provably safe and reliable active-learning-based multi-agent coverage algorithm. **Tech:** Python, C++, ROS, CasADi

Internship
Biotronik

Bülach, Switzerland
September 2018 - April 2019

- Developed statistical simulation and performed process capability analysis to optimize production workflows and validate test methods for quality control. **Tech:** Octave, LabView

EDUCATION

M.Sc. in Robotics, Systems and Control
Swiss Federal Institute of Technology (ETH), GPA: 5.82/6.00 (with distinction)

Zurich, Switzerland
2019 - 2021

Thesis: ▷ Developed a learning-based model predictive coverage algorithm with formal safety guarantees.

Semester Project: ▷ Designed a hierarchical quadratic programming controller with anti-reset windup for an exoskeletal rehabilitation robot.

B.Sc. in Mechanical Engineering, focus in Biomedical Engineering
Swiss Federal Institute of Technology (ETH), GPA: 5.46/6.00

Zurich, Switzerland
2015 - 2019

Thesis: ▷ Developed an ML-based optimization method for sensor number and placement in ultrasonic left ventricular volume estimation.

SELECTED PUBLICATIONS (FULL LIST ON MY GOOGLE SCHOLAR)

- [1] A. Scampicchio, L. F. Toso, **R. Rickenbach**, J. Anderson, and M. Zeilinger, “Physics-informed learning under mixing: How physical knowledge speeds up learning”, in *International Conference on Learning Representations*, 2026.
- [2] **R. Rickenbach**, A. Lahoud, E. Schaffernicht, M. Zeilinger, and J. A. Stork, “Zipmpc: Compressed context-dependent mpc cost via imitation learning”, in *Proceedings of The 9th Conference on Robot Learning*, PMLR, 2025.
- [3] **R. Rickenbach**, A. Lahr, and M. N. Zeilinger, “Inverse optimal control with constraint relaxation”, *IEEE Control Systems Letters*, 2025.
- [4] **R. Rickenbach**, J. Köhler, A. Scampicchio, M. N. Zeilinger, and A. Carron, “Active learning-based model predictive coverage control”, *IEEE Transactions on Automatic Control*, 2024.
- [5] **R. Rickenbach**, A. Scampicchio, and M. N. Zeilinger, “Inverse optimal control as an errors-in-variables problem”, in *6th annual learning for dynamics & control conference*, PMLR, 2024.

SUBMITTED PUBLICATIONS

- [6] **R. Rickenbach***, J. Trisovic*, A. Didier, J. Sieber, and M. N. Zeilinger, “Task-level insights from eigenvalues across sequence models”, *Neurips 2025 Dynafront Workshop, submitted to AAAI 2027*,
- [7] **R. Rickenbach**, B. Lee, R. Zurbrügg, C. A. Alonso, and M. N. Zeilinger, “Demonstrate: Zero-shot language to robotic control via multi-task demonstration learning”, *submitted to ICRA 2027*,

CURRENT RESEARCH

- Large Language Model Fine-Tuning using Inverse Reinforcement Learning
- Memory Capable Objective Regulation from Vision-Perceived Context
- Sampling-Based Diffusion Model Guidance to avoid Mode Averaging
- **Reviewer:** IEEE Transactions on Automatic Control, IEEE Transactions on Control of Network Systems, IEEE Transactions on Automation Science and Engineering, IEEE Control Systems Letters, IEEE Robotics and Automation Letters, Automatica, CoRL

SKILLS

- **Programming:** Python, MATLAB, C++, SQL
- **ML & scientific computing:** PyTorch, JAX, Hugging Face, NumPy, SciPy, Weights & Biases
- **Robotics & control:** ROS, CasADi, PandaGym
- **Tools:** Linux, Git, Docker, $\text{\LaTeX}$

LANGUAGES

- **English:** fluent, C1
- Cambridge Advanced
- **French:** intermediate, B2
- DELF-Zertifikat
- **German:** mother tongue

AWARDS

- SGA advancement award 2022
- Recipient of NCCR Robotics Equal Opportunities Committee grant 2022
- Outstanding Teaching Assistant Award 2020

MISCELLANEOUS

- **Teaching Experience** — Designed and taught graduate- and undergraduate-level ETH courses since 2017; supervised master’s and bachelor’s theses and semester projects since 2022.
- **Outreach and Community Engagement** — Contributing to STEM outreach initiatives supporting girls and young women; organizing social and recreational events for institute members.
- **Scholarship – Femtec Career Building Program** (2018–2019) — Selected for this program, participating in workshops on professional skills, project planning, and innovation management; completed an innovation project at Porsche AG on manufacturing simulation quality for cast components.